

Input

TR locus

A T T A T T ...

C G G C G G ...

A G A G A G ...

T C C T T C ...

...

Feature Selection

Gene feature

Gene1

5'UTR

exon

Gene2

A T T A T T ...

exon

Intron

3'UTR

Sequence feature

flanking

A T T A T T ...

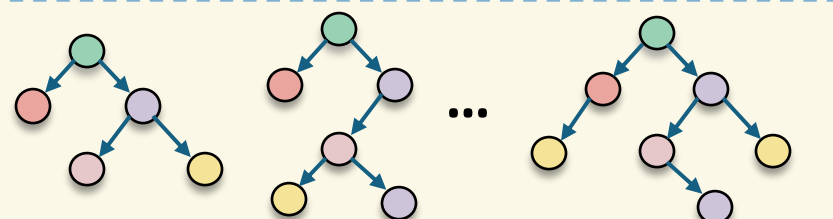
Ref

TR

G C C ...

Ensemble learning framework

CatBoost Models



$$y = \sum_{t=1}^T \gamma_t h_t(x)$$

y_{i1} y_{i2} ... y_{in}

Stacking

$$y = \text{Meta}(y_1, y_2, \dots, y_n)$$

Logistic Regression

Output

pathogenic

benign